Chapter 4 Maxwell Equations

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4.1 Maxwell Equations

So far, we have considered only static fields with no time variation. In this chapter, we explore the effects of introducing time-varying fields into Maxwell equations.

4.1.1 Faraday's Law

When there exists time varying magnetic field, we modify the curl equation for electric field in the electrostatic case according to Faraday's law as

$$\begin{cases} \oint_C \vec{E} \cdot d\vec{l} = -\dfrac{d}{dt}\displaystyle\int_S \vec{B} \cdot d\vec{s} \\ \vec{\nabla} \times \vec{E} = -\partial_t \vec{B} \end{cases} \tag{4.1.1}$$

It is noted that $\vec{E}$ is no longer conservative. Define $\mathcal{V}_{emf} = \oint_C \vec{E} \cdot d\vec{l}$ as the electromotive force (emf) induced in the circuit and $\Phi = \int_S \vec{B} \cdot d\vec{s}$ as the magnetic flux passing the surface, then we get

$$\mathcal{V}_{emf} = -\frac{d\Phi}{dt} \tag{4.1.2}$$

which is known as the Lenz's law. The minus sign in (4.1.2) indicates that the induced emf tends to "counter-react" the change of magnetic flux. It is

noted that for $\mathcal{V}_{emf}$ to be non-zero, either $\vec{B}$ and/or $\vec{s}$ must be time-varying. Thus, three cases can be considered:

i. stationary $\vec{s}$ with time-varying $\vec{B}$
ii. time-varying $\vec{s}$ with stationary $\vec{B}$
iii. time-varying $\vec{s}$ with time-varying $\vec{B}$

We will cover these three situations with three following examples.

A. Case i: stationary $\vec{s}$ with time-varying $\vec{B}$

□ **<u>Example 4-1</u>** <u>(Cheng 7-1)</u>

A circular loop with N turns of conducting wire is positioned in the xy-plane, centered at the origin. It is placed in a magnetic field given by $\vec{B} = \hat{z}B_0 \cos\left(\frac{\pi r}{2b}\right)\sin(\omega t)$ where b is the radius and ω is the angular frequency. Determine the emf induced in the loop.

Solution

$$\Phi = \int_S \vec{B} \cdot d\vec{s} = \int_0^{2\pi}\int_0^b B_0 \cos\left(\frac{\pi r}{2b}\right)\sin(\omega t) r dr d\theta$$
$$= \frac{8b^2}{\pi}\left(\frac{\pi}{2} - 1\right) B_0 \sin(\omega t)$$
$$\mathcal{V}_{emf} = -N\frac{d\Phi}{dt} = \frac{8Nb^2}{\pi}\left(\frac{\pi}{2} - 1\right) B_0 \sin(\omega t)$$

Note: $\int x \cos x \, dx = \int x d \sin x = x \sin x + \cos x$.

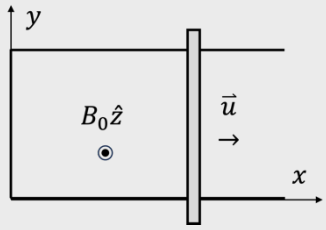
B. Case ii: time-varying $\vec{s}$ with stationary $\vec{B}$

□ **<u>Example 4-2</u>** <u>(Cheng 7-2)</u>

A metal bar moves with a constant velocity $\vec{u}$ along a pair of conducting rails in a uniform magnetic field $B_0\hat{z}$ as shown in the figure. Determine the emf in the loop.

Solution

$$\mathcal{V}_{emf} = -\frac{d\Phi}{dt} = -\frac{d}{dt}(B_0 \cdot x \cdot h)$$
$$= -B_0 h \frac{dx}{dt} = -B_0 h u$$



C. Case iii: time-varying $\vec{s}$ with time-varying $\vec{B}$

□ **<u>Example 4-3</u>** (Cheng 7-4)

A rectangular conducting loop of dimensions $h \times w$ with its rotation axis aligned with $\hat{x}$ is placed in a time-varying magnetic field $\vec{B} = B_0 \sin(\omega t)\,\hat{y}$. The normal to the loop initially forms an angle α with $\hat{y}$.

Determine the induced emf in the loop:
(a) when the loop is stationary, and
(b) when the loop rotates about the x-axis with an angular velocity ω.

Solution

(a)

1. $\Phi = \int \vec{\mathrm{B}} \cdot d\vec{s} = B_0 \sin(\omega t)\,\hat{y} \cdot hw\,\hat{n} = B_0\, hw \sin(\omega t) \cos\alpha$
2. $\mathcal{V}_{emf} = -\frac{d\Phi}{dt} = -B_0\, hw\omega \cos(\omega t) \cos\alpha$

(b)

1. $\alpha = \omega t$, $\Phi = B_0\, hw \sin(\omega t) \cos(\omega t)$
2. $\mathcal{V}_{emf} = -\frac{d\Phi}{dt} = -B_0\, hw\omega \cos(2\omega t)$

We have incorporated the time-varying magnetic field into Maxwell's equations via Faraday's law (4.1.1), which states that a changing magnetic field induces an electric field, addressing electromagnetic induction. To fully describe time-dependent fields, we also need to include the effects of a time-varying electric field to ensure the correct interaction between the electric and magnetic fields. This is done by modifying Ampère's law based on the continuity equation, completing Maxwell equations for both static and time-varying fields.

4.1.2 Continuity Equation

From circuit theory, consider a fixed volume V enclosed by a surface S, we know that at any point, the current $\mathcal{I}$ is related to the charge Q with

$$\mathcal{I} = -\frac{dQ}{dt} \tag{4.1.3}$$

(4.1.3) shows that a decrease in charge density corresponds to an outflowing current, and vice versa. From a field point of view, we have $\mathcal{I} = \oint_S \vec{J} \cdot d\vec{s}$ and $Q = \int_V \rho dv$. Substituting back into (4.1.3), we get the continuity equation in integral form

$$\oint_S \vec{J} \cdot d\vec{s} = \int_V \vec{\nabla} \cdot \vec{J} dv = -\frac{d \int_V \rho dv}{dt} = \int_V -\partial_t \rho dv \tag{4.1.4}$$

and by applying the divergence theorem, we get the continuity equation in differential form

$$\vec{\nabla} \cdot \vec{J} = -\partial_t \rho \tag{4.1.5}$$

Note that in (4.1.5), the surface current density accounts for the time-varying change in charge density, in contrast to the magnetostatic case where the charge distribution is constant. In other words we have $\vec{\nabla} \cdot \vec{J} = 0$ for magnetostatics.

4.1.3 Ampère-Maxwell's Law

Ampère's law introduced for the magnetostatic case $\vec{\nabla} \times \vec{H} = \vec{J}$ should be revised since $\vec{\nabla} \cdot \vec{\nabla} \times \vec{H} \equiv 0$ but $\vec{\nabla} \cdot \vec{J} = -\partial_t \rho$. An additional term $\partial_t \vec{D}$ (displacement current) is introduced by Maxwell to the Ampère's law to fix this inequality

$$\vec{\nabla} \times \vec{H} = \vec{J} + \partial_t \vec{D} \tag{4.1.6}$$

This is the Ampère-Maxwell's law. Notice that the divergence of right-hand side is $\vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \partial_t \vec{D} = \vec{\nabla} \cdot \vec{J} + \partial_t \vec{\nabla} \cdot \vec{D} = \vec{\nabla} \cdot \vec{J} + \partial_t \rho \equiv 0$, which now satisfy the constraint of $\vec{\nabla} \cdot \vec{\nabla} \times \vec{H} \equiv 0$. With Faraday's law and the inclusion of the term $\partial_t \vec{D}$ in the Ampère–Maxwell law, Maxwell equations are completed to account for time-varying fields. Next, we will introduce the full set of Maxwell equations in both integral and differential form.

4.1.4 Integral and Differential Form

By applying what we have learned thus far, we are now able to write the full set of Maxwell equations in integral form

$$\oint_C \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{s} \tag{4.1.7a}$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{s} + \frac{d}{dt}\int_S \vec{D} \cdot d\vec{s} \tag{4.1.7b}$$

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \, dv \tag{4.1.7c}$$

$$\oint_S \vec{B} \cdot d\vec{s} = 0 \tag{4.1.7d}$$

and the differential form

$$\vec{\nabla} \times \vec{E} = -\partial_t \vec{B} \tag{4.1.8a}$$

$$\vec{\nabla} \times \vec{H} = \partial_t \vec{D} + \vec{J} \tag{4.1.8b}$$

$$\vec{\nabla} \cdot \vec{D} = \rho \tag{4.1.8c}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \tag{4.1.8d}$$

Note that the Maxwell equations consist of two curl equations and two divergence equations. The divergence equations (c–d) describe how fields originate from sources: Gauss's law for electricity relates the divergence of the electric field to the local charge density, while Gauss's law for magnetism states that the divergence of the magnetic field is zero, indicating the absence of magnetic monopoles. The curl equations (a–b) govern the dynamic interplay between electric and magnetic fields. Faraday's law (a) states that a time-varying magnetic field induces a circulating electric field. The Ampère–Maxwell law (b) shows that both conduction current and a time-varying electric displacement field act as sources of a circulating magnetic field. These equations highlight the mutual generation of electric and magnetic fields in response to time variations, demonstrating their intrinsic coupling in the dynamic regime.

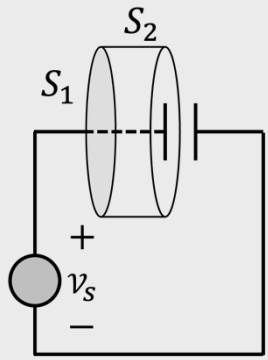
Together, the four equations form a complete framework for describing the behavior of electromagnetic fields in both static and time-varying situations.

□ **Example 4-4** (Cheng 7-5)

An AC voltage source with the voltage $\mathcal{V}_c = \mathcal{V}_0 \sin(\omega t)$, is connected across a parallel-plate capacitor C.

(a) Verify that the displacement current in the capacitor equals the conduction current in the wires.
(b) Determine the magnetic field intensity at a distance r from the wire.

Solution



(a)

1. $\mathcal{I}_c = C\frac{d\mathcal{V}_c}{dt} = \omega C\mathcal{V}_0 \cos(\omega t)$
2. $E = \frac{\mathcal{V}_c}{d}$, $D = \epsilon_0 E = \frac{\epsilon_0 \mathcal{V}_0 \sin(\omega t)}{d}$
3. $\partial_t D = \frac{\omega \epsilon_0 \mathcal{V}_0 \cos(\omega t)}{d}$
4. $\mathcal{I}_d = \int \partial_t \vec{D} \cdot d\vec{s} = \omega \frac{\epsilon_0 A}{d} \mathcal{V}_0 \cos(\omega t) = \mathcal{I}_c$

(b)

Either surface S_1 or S_2 can be chose

$$H = \frac{\mathcal{I}_c}{2\pi r} = \frac{\mathcal{I}_d}{2\pi r} = \frac{\omega C\mathcal{V}_0 \cos(\omega t)}{2\pi r}$$

4.2 Electromagnetic Matching Conditions

The matching conditions for time-varying fields are identical to those derived for static fields under the static assumption. Since the integration of the finite time varying fields over an infinitesimal area or volume is zero, the derivation follows the same procedure in both cases. We will directly present the final results:

$$\hat{n} \times \left(\vec{E}_1 - \vec{E}_2\right) = 0 \tag{4.2.1a}$$

$$\hat{n} \times \left(\vec{H}_1 - \vec{H}_2\right) = \vec{J}_s \tag{4.2.1b}$$

$$\hat{n} \cdot \left(\vec{D}_1 - \vec{D}_2\right) = \rho_s \tag{4.2.1c}$$

$$\hat{n} \cdot \left(\vec{B}_1 - \vec{B}_2\right) = 0 \tag{4.2.1d}$$

where $\hat{n}$ is the unit vector pointing from medium 2 to medium 1. The following points are the general statements:

i. The tangential component of $\vec{E}$ is continuous across an interface.
ii. The tangential component of $\vec{H}$ is discontinuous at an interface with a surface current, as given by (4.2.1b).
iii. The normal component of $\vec{D}$ is discontinuous at an interface with a surface charge, as given by (4.2.1c).
iv. The normal component of $\vec{B}$ is continuous across an interface.

(4.2.1a-d) are general cases that are applicable for any medium. Let us consider the follwing special cases: (a) For surface current $\vec{J}_s$ to exist, either medium 1 or 2 should be PEC ($\sigma \to \infty$), and inside PEC, the fields are zero. (b) When both media are lossless dielectrics ($\sigma = 0$), in most cases the sources term at the right-hand side of (4.2.1b-c) vanishes, that is, no free surface charge density of surface currents on the boundary between two dielectrics.

4.3 Potential Functions

To solve Maxwell equations for time-varying fields, we use a combination of the scalar potential from electrostatics and the vector potential from magnetostatics.

First, as in magnetostatics, we use the magnetic vector potential

$$\begin{aligned} \vec{B} &= \vec{\nabla} \times \vec{A} \\ \vec{H} &= \frac{1}{\mu}\vec{\nabla} \times \vec{A} \end{aligned} \tag{4.3.1}$$

From (4.1.8a), where $\vec{E}$ can be expressed as

$$\vec{E} = -\partial_t \vec{A} - \vec{\nabla}\mathcal{V} \tag{4.3.2}$$

Substituting (4.3.1) and (4.3.2) into (4.1.8b)

$$\vec{\nabla} \times \vec{\nabla} \times \vec{A} + \mu\epsilon\partial_t^2 \vec{A} + \mu\epsilon\partial_t \vec{\nabla}\mathcal{V} = \mu\vec{J} \tag{4.3.3}$$

Use the identity $\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$, we get

$$\nabla^2 \vec{A} - \mu\epsilon\partial_t^2 \vec{A} = -\mu\vec{J} + \vec{\nabla}(\vec{\nabla} \cdot \vec{A} + \mu\epsilon\partial_t \mathcal{V}) \tag{4.3.4}$$

Note that only the curl of the magnetic vector potential is defined. According to Helmholtz's theorem, we have the freedom to specify its divergence:

$$\vec{\nabla} \cdot \vec{A} + \mu\epsilon\partial_t \mathcal{V} = 0 \tag{4.3.5}$$

This is known as the Lorentz gauge. Then, we arrive at the inhomogeneous wave equation for the scalar potential $\mathcal{V}$

$$\nabla^2 \mathcal{V} - \mu\epsilon\partial_t^2 \mathcal{V} = -\frac{\rho}{\epsilon} \tag{4.3.6}$$

and the vector potential $\vec{A}$

$$\nabla^2 \vec{A} - \mu\epsilon\partial_t^2 \vec{A} = -\mu\vec{J} \tag{4.3.7}$$

Now $\vec{A}$ and $\mathcal{V}$ are uncoupled. Given charge and current distributions, we first solve the inhomogeneous wave equations (4.3.6-7) to determine the potentials $\vec{A}$ and $\mathcal{V}$. Once $\vec{A}$ and $\mathcal{V}$ are obtained, the electric and magnetic fields can be computed using (4.3.1-2) through differentiation.

4.4 Wave Equations

For source-free Maxwell equations, one can get the homogenous vector wave functions for both electric and magnetic field

$$\nabla^2 \vec{E} - \mu\epsilon\partial_t^2 \vec{E} = 0 \tag{4.4.1a}$$

$$\nabla^2 \vec{H} - \mu\epsilon\partial_t^2 \vec{H} = 0 \tag{4.4.1b}$$

Suppose we have a wave propagating in the *x*-direction, and the fields have no variations in *y* and *z* directions, then every Cartesian field component *f*

in (4.4.1) is the solution of the one-dimensional homogeneous wave equation

$$\partial_x^2 f - \mu\epsilon\partial_t^2 f = 0 \tag{4.4.2}$$

and the solution takes the form of

$$f = f^+\left(t - \frac{x}{v}\right) + f^-\left(t + \frac{x}{v}\right) \tag{4.4.3}$$

where f^+ denotes the wave propagating in the positive x-direction, and f^- denotes the wave propagating in the negative x-direction. $v = \frac{1}{\sqrt{\mu\epsilon}}$ is the velocity of the wave.

For inhomogeneous scalar wave equations (4.3.6), analogous to the static case (2.1.24), the solution is[1]

$$\mathcal{V}(R,t) = \frac{1}{4\pi\epsilon}\int_{V'} \frac{\rho\left(t - \frac{R}{u}\right)}{R} dv' \tag{4.4.4}$$

where R is the distance between the source point and the observation point. Equation (4.4.2) shows that the scalar potential at a distance R depends on the charge density at an earlier time $t - \frac{R}{u}$, accounting for the time delay in propagation. This is why $\mathcal{V}(R,t)$ is called the retarded scalar potential. A function of $t + \frac{R}{u}$ would be unphysical, as it implies effects occurring before their cause.

Similarly, solving the inhomogeneous wave equation (4.3.7) for the vector potential $\vec{A}$ follows the same approach. The solution also yields retarded vector potential

$$\vec{A}\,(R,t) = \frac{\mu}{4\pi}\int_{V'} \frac{\vec{J}\left(t - \frac{R}{u}\right)}{R} dv' \tag{4.4.5}$$

Wave equations—and Maxwell equations in general—are often challenging to solve directly in the time domain. To simplify the analysis, it is common to transform the problem into the frequency domain. In the

[1] Not proven here due to the depth of the topic. The interested reader should refer to the method of Green function.

next section, we introduce the phasor technique, a method that represents time-varying fields as complex exponentials. This approach converts differential equations with time derivatives into algebraic equations, greatly facilitating the analysis of steady-state sinusoidal behavior in linear systems.

4.5 Time-Harmonic Fields

In many practical applications of electromagnetics, fields and sources vary sinusoidally with time. Such time-harmonic fields simplify the analysis of Maxwell equations by allowing solutions to be expressed in terms of complex exponentials. A time-harmonic field can be represented as

$$\vec{E}(\vec{r},t) = \Re\left[\vec{E}(\vec{r})e^{i\omega t}\right] \tag{4.5.1}$$

where $\vec{F}(\vec{r},t)$ is the real-valued time-domain field, $\vec{F}(\vec{r})$ is its complex-valued phasor, ω is the angular frequency, i is the imaginary unit, and $\Re(\cdot)$ denotes taking the real part. Note that the same notation is used for both time-domain fields and phasors; the context should clarify which representation is being used.

By substituting time-harmonic fields into Maxwell equations, time derivatives ∂_t are replaced by multiplication with $i\omega$, reducing the equations to algebraic forms in the frequency domain.

$$\vec{\nabla} \times \vec{E} = -i\omega\vec{B} \tag{4.5.2a}$$

$$\vec{\nabla} \times \vec{H} = i\omega\vec{D} + \vec{J} \tag{4.5.2b}$$

$$\vec{\nabla} \cdot \vec{D} = \rho \tag{4.5.2c}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \tag{4.5.2d}$$

Note that when constitutive relations are used in (4.5.2b)

$$\vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E} + \sigma\vec{E} = i\omega\epsilon_c\vec{E} \tag{4.5.3}$$

where

$$\epsilon_c = \epsilon - i\frac{\sigma}{\omega} = \epsilon' - i\epsilon'' \tag{4.5.4}$$

is defined as the complex permeability, where ϵ' is the real part of ϵ_c, and ϵ'' is the negative imaginary part of ϵ_c. In general, ϵ can also be a complex number. The loss tangent of a material is defined as the negative ratio of the imaginary part to the real part of its complex permeability. For real ϵ, the loss tangent is expressed as

$$\tan \delta_c = \frac{\sigma}{\omega\epsilon} \tag{4.5.5}$$

If, $\sigma \gg \omega\epsilon$, the material is characterized as good conductors, If, $\sigma \ll \omega\epsilon$, the material is characterized as good insulators. It is noted that the comparison is relative to the frequency of interest.

The wave equation for the scalar potential $\mathcal{V}$ and its solution in frequency domain is

$$\nabla^2\mathcal{V} - k^2\mathcal{V} = -\frac{\rho}{\epsilon} \tag{4.5.5a}$$

$$\mathcal{V} = \frac{1}{4\pi\epsilon}\int_{V'} \frac{\rho e^{-ikR}}{R} dv' \tag{4.5.5b}$$

where $k = \omega\sqrt{\mu\epsilon}$ is the wavenumber. The vector potential $\vec{A}$ and its solution in frequency domain is

$$\nabla^2\vec{A} + k^2\vec{A} = -\mu\vec{J} \tag{4.5.6a}$$

$$\vec{A} = \frac{\mu}{4\pi}\int_{V'} \frac{\vec{J}e^{-ikR}}{R} dv' \tag{4.5.6b}$$

It is evident from a direct comparison between the equations in the frequency domain and those in the time domain that the phasor transformation simplifies the time derivatives into algebraic multiplications by $i\omega$, making the analysis of time-harmonic fields significantly more tractable.

4.6 Poynting's Theorem

Poynting's theorem is an energy conservation law for electromagnetic fields. It describes how electromagnetic energy flows through space and how it is transferred between fields and matter. The theorem introduces the Poynting vector, which represents the power per unit area carried by the electromagnetic field, and relates the rate of energy decrease in a volume

to the net outward energy flux and the work done on charges within the volume.

From (4.1.8a-b) and the vector identity $\vec{\nabla}\cdot\left(\vec{A}\times\vec{B}\right)=\vec{A}\cdot\left(\vec{\nabla}\times\vec{B}\right)-\vec{A}\cdot\left(\vec{\nabla}\times\vec{B}\right)$, we have

$$\begin{aligned}\vec{\nabla}\cdot\left(\vec{E}\times\vec{H}\right)&=\vec{H}\cdot\left(\vec{\nabla}\times\vec{E}\right)-\vec{E}\cdot\left(\vec{\nabla}\times\vec{H}\right)\\&=-\vec{H}\cdot\partial_t\vec{B}-\vec{E}\cdot\partial_t\vec{D}-\vec{E}\cdot\vec{J}\end{aligned}\tag{4.6.1}$$

In a simple medium (linear, homogeneous and isotropic), we can simplify the terms in the right-hand side with

$$\vec{H}\cdot\partial_t\vec{B}=\vec{H}\cdot\partial_t\mu\vec{H}=\partial_t\left(\frac{\mu}{2}\left|\vec{H}\right|^2\right)$$

$$\vec{E}\cdot\partial_t\vec{D}=\vec{E}\cdot\partial_t\epsilon\vec{E}=\partial_t\left(\frac{\epsilon}{2}\left|\vec{E}\right|^2\right)$$

$$\vec{E}\cdot\vec{J}=\sigma\left|\vec{E}\right|^2$$

Thus, (4.6.1) becomes

$$\vec{\nabla}\cdot\left(\vec{E}\times\vec{H}\right)=-\partial_t\left(\frac{\mu}{2}\left|\vec{H}\right|^2+\frac{\epsilon}{2}\left|\vec{E}\right|^2\right)-\sigma\left|\vec{E}\right|^2\tag{4.6.2}$$

Integrating (4.6.2) over a volume and apply divergence theorem, we get

$$\begin{aligned}&\oint_S\left(\vec{E}\times\vec{H}\right)\cdot d\vec{s}\\&\qquad=-\partial_t\int_V\left(\frac{\mu}{2}\left|\vec{H}\right|^2+\frac{\epsilon}{2}\left|\vec{E}\right|^2\right)dv-\int_V\sigma\left|\vec{E}\right|^2dv\end{aligned}\tag{4.6.3}$$

The first two terms on the right side of (4.6.3) represent the time rate of change of energy stored in the electric and magnetic fields, respectively, while the last term accounts for the ohmic power dissipation due to conduction currents in the presence of an electric field. Thus, (4.6.3) describes the rate of decrease in stored electromagnetic energy, minus the power lost as heat, which must equal the power flowing out of the volume through its surface. This leads to the definition of the Poynting vector

$$\vec{S}=\vec{E}\times\vec{H}\tag{4.6.4}$$

which represents the electromagnetic power flow per unit area. The surface integral of $\vec{S}$ over a closed surface gives the total power leaving the enclosed volume, a statement known as Poynting's theorem.

□ **<u>Example 4-5</u>** <u>(Cheng 8-7)</u>

Determine the Poynting vector on the surface of a long, straight conducting wire with radius b and conductivity σ, carrying a direct current I. Verify Poynting's theorem.

Solution

$$\vec{J} = \frac{I}{\pi b^2}\hat{z}$$

$$\vec{E} = \frac{\vec{J}}{\sigma} = \frac{I}{\sigma\pi b^2}\hat{z}$$

On the surface of the wire,

$$\vec{H} = \frac{I}{2\pi b}\hat{\phi}$$

Thus,

$$\vec{S} = \vec{E} \times \vec{H} = -\hat{r}\frac{I^2}{2\sigma\pi^2 b^3}$$

We integrate the Poynting vector over cylinder with a segment length l

$$-\oint_S \vec{S} \cdot d\,\vec{s} = \frac{I^2}{2\sigma\pi^2 b^3} 2\pi b l = I^2\left(\frac{l}{\sigma\pi b^2}\right) = I^2 R$$

which is equal to the Ohmic loss.

For time-harmonic cases, from (4.6.4)

$$\vec{S}(\vec{r},t) = \Re\left[\vec{E}(\vec{r})e^{i\omega t}\right] \times \Re\left[\vec{H}(\vec{r})e^{i\omega t}\right]$$

$$= \frac{\vec{E}(\vec{r})e^{i\omega t} + \vec{E}^*(\vec{r})e^{-i\omega t}}{2} \times \frac{\vec{H}(\vec{r})e^{i\omega t} + \vec{H}^*(\vec{r})e^{-i\omega t}}{2} \qquad (4.6.5)$$

$$= \frac{1}{2}\Re\left[\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r}) + \vec{E}(\vec{r}) \times \vec{H}(\vec{r})e^{i2\omega t}\right]$$

In most cases the time-average value is more important than the instantaneous value, the time-average Poynting vector can be obtained by integrating the instantneous Poynting vector over a period T and divided by T

$$\vec{S}_{av}(\vec{r},t) = \frac{1}{T}\int_0^T \vec{S}(\vec{r},t) = \frac{1}{2}\Re\left[\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})\right] \tag{4.6.6}$$

Since the term $\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})$ in (4.5.5) is constant over time and the average of $\vec{E}(\vec{r}) \times \vec{H}(\vec{r})e^{i2\omega t}$ is zero over a period.

□ **<u>Example 4-6</u>** <u>(Cheng 8-8)</u>

The far field of a short current element $\mathcal{I}dl\hat{z}$ located at the origin can be expressed as

$$\vec{E}(r,\theta) = i\left(\frac{60\pi\mathcal{I}dl\sin\theta}{\lambda}\right)\frac{e^{-ikr}}{r}\hat{\theta}$$

$$\vec{H}(r,\theta) = i\left(\frac{\mathcal{I}dl\sin\theta}{2\lambda}\right)\frac{e^{-ikr}}{r}\hat{\phi}$$

Find (a) the instantaneous Poynting vector and (b) the total average power radiate by the current element.

Solution

(a)

$$\begin{aligned}\vec{S}(r,\theta,t) &= \Re\left[\vec{E}(r,\theta)e^{i\omega t}\right] \times \Re\left[\vec{H}(r,\theta)e^{i\omega t}\right] \\ &= \hat{r}\,30\pi\left(\frac{\mathcal{I}dl\sin\theta}{r\lambda}\right)^2 \sin^2(\omega t - kr) \\ &= \hat{r}\,30\pi\left(\frac{\mathcal{I}dl\sin\theta}{r\lambda}\right)^2 [1 - 2\cos(\omega t - kr)\,]\end{aligned}$$

(b)

$$\vec{S}_{av}(r,\theta) = \frac{1}{2}\Re\left[\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})\right] = \hat{r}\,15\pi\left(\frac{\mathcal{I}dl\sin\theta}{r\lambda}\right)^2$$

Total average power:

$$P_{av} = \oint_S \vec{S}_{av}(r,\theta) \cdot d\vec{s} = \int_0^{2\pi} \int_0^{\pi} \left[15\pi \left(\frac{\mathcal{I} dl \sin\theta}{r\lambda} \right)^2 \right] r^2 \sin\theta \, d\theta d\phi$$
$$= 40 \left(\frac{\mathcal{I} dl}{\lambda} \right)$$